SUMMER REVIEW FOR AP CALCULUS

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Give the domain of the function.

1) $f(x) = \sqrt{3 - x}$

Evaluate the function.

2) Find $f(a + 2)$ when $f(x) = x^2 + 5$.

Find the average rate of change for the function over the given interval.

3) $f(x) = \dfrac{3}{x + 2}$ between $x = 1$ and $x = 4$

Determine if the given function is even, odd, or neither.

4) $f(x) = x^3 + x^2 + 1$

Find the requested function value.

5)

If $f(x) = \begin{cases} x^3 & \text{if } x < 0 \\ 4x + 2 & \text{if } x \geq 0 \end{cases}$ find $f(-4)$

Tell whether the graph of the function opens upward or downward and whether the graph is wider, narrower, or the same as $f(x) = x^2$.

6) $f(x) = \dfrac{1}{8}x^2 - 2$

Find the requested value.

7) $f(x) = 9x + 9$, $g(x) = 6x - 8$
Find fg.

Find the indicated composite for the pair of functions.

8) $(f \circ g)(x)$: $f(x) = \dfrac{2}{x + 2}$, $g(x) = \dfrac{5}{4x}$

Solve the problem.

9) The owner of a video store has determined that the profits P of the store are approximately given by $P = -x^2 + 130x + 69$, where x is the number of videos rented daily. Find the maximum profit to the nearest dollar.

Find the x-intercepts and y-intercepts (if they exist).

10) $f(x) = x^2 - 5$

For the polynomial, list each real zero and its multiplicity. Determine whether the graph crosses or touches the x-axis at each x -intercept.

11) $f(x) = 4(x - 4)(x - 2)^2$

Decide whether the graph has an asymptote, a hole, or neither at x = t.

12) $\dfrac{t}{x - t}$

Give the equation of the specified asymptote(s).

13) Vertical asymptote(s): $f(x) = \dfrac{x - 5}{x^2 - 36}$

Answer the question.

14) Is there y-axis symmetry for the rational function $f(x) = \dfrac{-4x^2}{-2x^4 - 20}$?

If the following defines a one-to-one function, find the inverse.

15) $4y = 5x - 6$

Decide whether or not the functions are inverses of each other.

16) $f(x) = 5 - 3x$; $g(x) = -\dfrac{x}{3}(x - 5)$

Approximate each value using a calculator. Express answer rounded to three decimal places.

17) $3.4^{3.07}$

Solve the equation.

18) $2^{(1 + 2x)} = 32$

Convert to logarithmic form.

19) $e^x = 17$

Solve the equation.

20) $3^{-x} = \dfrac{1}{81}$

Solve the problem.

21) $\log_3 x = -2$

22) The formula $D = 7e^{-0.04h}$ can be used to find the number of milligrams D of a certain drug in a patient's bloodstream h hours after the drug has been given. When the number of milligrams reaches 4, the drug is to be given again. What is the time between injections?

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Write as the sum and/or difference of logs. Do not use exponents.

23) $\log_{16} \frac{5}{9}$

Use a calculator and the base conversion formula to find the logarithm, correct to three decimal places.

24) $\log_{2.8} 244$

Solve the equation.

25) $\log (4 + x) - \log (x - 5) = \log 2$

Convert the degree measurement to radians. Express answer as multiple of π.

26) 144°

Convert the radian measure to degrees. (Round to the nearest hundredth when necessary)

27) $\frac{11\pi}{4}$

Find the exact value of the requested trigonometric function of θ.

28) $\sin \theta = -\frac{2}{5}$ and $\tan \theta > 0$

Find $\sec \theta$.

If s denotes the length of the arc of a circle of radius r subtended by a central angle θ, find the missing quantity.

29) $r = \frac{1}{5}$ feet, $s = 4$ feet, $\theta = ?$

In the problem, t is a real number and P=(x,y) is the point on the unit circle that corresponds to t. Find the exact value of the given trigonometric function.

30) $\left(-\frac{\sqrt{55}}{8}, \frac{3}{8}\right)$; find $\cos t$

Find the exact value of the expression. Do not use a calculator.

31) $\sin \frac{11\pi}{3}$

Use the fact that the trigonometric functions are periodic to find the exact value of the expression.

32) $\cot \frac{13\pi}{4}$

Find the amplitude or period as requested.

33) Find the period of $y = -3 \sin (6x + \frac{\pi}{2})$.

Name the quadrant in which the angle θ lies.

34) $\cot \theta < 0$ and $\cos \theta > 0$

Solve.

35) For what numbers x, $-2\pi \le x \le 2\pi$, does the graph of $y = \sec x$ have vertical asymptotes.?

Find the value of the expression.

36) $\cos^{-1}\left(-\frac{\sqrt{2}}{2}\right)$

Complete the identity.

37) $\tan \theta(\cot \theta - \cos \theta) = ?$

38) $\frac{1}{\cos^2\theta} - \frac{1}{\cot^2\theta} = ?$

Find the exact value by using a sum or difference identity.

39) $\sin \frac{\pi}{12}$

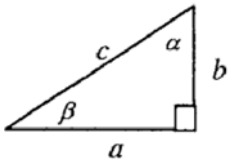
Use trigonometric identities to find the exact value.

40) $\sin 25^\circ \cos 35^\circ + \cos 25^\circ \sin 35^\circ$

Solve the equation for solutions in the interval $[0, 2\pi)$.

41) $2 \cos \theta + 1 = 0$

42) Solve the right triangle given that $\alpha = 35^\circ$ and $a = 8$.



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1) Answer: $x \le 3$

2) Answer: $a^2 + 4a + 9$

3) Answer: $-\frac{1}{6}$

4) Answer: Neither

5) Answer: –64

6) Answer: Upward, wider

7) Answer: $54x^2 - 18x - 72$

8) Answer: $\frac{8x}{5 + 8x}$

9) Answer: $4294

10) Answer: x–intercepts $(\sqrt{5}, 0)$ and $(-\sqrt{5}, 0)$; y–intercept (0,–5)

11) Answer: 4, multiplicity 1, crosses x–axis; 2, multiplicity 2, touches x–axis

12) Answer: Asymptote

13) Answer: $x = 6$, $x = -6$

14) Answer: Yes

15) Answer: $f^{-1}(x) = \frac{4x + 6}{5}$

16) Answer: No

17) Answer: 42.819

18) Answer: {2}

19) Answer: $\ln 17 = x$

20) Answer: {4}

21) Answer: $\{\frac{1}{9}\}$

22) Answer: 13.99 hrs

23) Answer: $\log_{16} 5 - \log_{16} 9$

24) Answer: 5.339

25) Answer: {14}

26) Answer: $\frac{4\pi}{5}$

27) Answer: 495°

28) Answer: $-\frac{5\sqrt{21}}{21}$

29) Answer: 20 radians

30) Answer: $-\frac{\sqrt{55}}{8}$

31) Answer: $-\frac{\sqrt{3}}{2}$

32) Answer: 1

33) Answer: $\frac{\pi}{3}$

34) Answer: Quadrant IV

35) Answer: $-\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}$

36) Answer: $\frac{3\pi}{4}$

37) Answer: $1 - \sin\theta$

38) Answer: 1

39) Answer: $\frac{\sqrt{2}(\sqrt{3} - 1)}{4}$

40) Answer: $\frac{\sqrt{3}}{2}$

41) Answer: $\frac{2\pi}{3}, \frac{4\pi}{3}$

42) Answer: $\beta = 55°$
$b = 11.43$
$c = 13.95$

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